

DS	Session	Sorting	Question Information	Level 1	Challenge 19
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Problem Description:

Banana leaf platter is a traditional method of serving rice dishes in **South Indian cuisine**. Due to the migration of South Indians, banana leaf rice can also be found in areas with significant ethnic South Indian diaspora such as **Malaysia** and **Singapore**.

Irfan is a banana leaf sales person.
he has N stacks of banana leaves.

Each stack contains K leaves.

Each leaf has a positive beauty value, describing how attractive it looks.

Irfan would like to take exactly P leaves to use for lunch today. If he would like to take a leaf in a stack, he must also take all of the leaves above it in that stack as well.

Help Irfan pick the P leaves that would maximize the total sum of attractive values.

Constraints:

$1 \leq T \leq 100.$

$1 \leq K \leq 30.$

$1 \leq P \leq N * K.$

$1 \leq N \leq 50.$

Input Format:

The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the three integers N, K and

P. Then, N lines follow. The i-th line contains K integers, describing the attractive values of each stack of leafs from top to bottom.

Output Format:

Print the output in a separate line contains the maximum total sum of attractive values that Irfan could pick.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
2 4 5
10 10 100 30
80 50 10 50
3 2 3
80 80
20 20
15 50
```

EXPECTED OUTPUT

```
250
180
```

Test Case 2

INPUT (STDIN)

```
2
2 4 5
20 20 70 30
60 30 10 50
3 2 3
60 70
15 40
20 10
```

EXPECTED OUTPUT

```
200
150
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int max(int a,int b)
```

Test Case 2

KEYWORD

```
for(int i = 0;i < n;i++)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

380

Test Case 3

NLOC

47

```
Sorting9.c +
1 #include <stdio.h>
2 #include <string.h>
3
4 int max(int a, int b) {
5     return (a > b) ? a : b;
6 }
7
8 int main() {
9     int t;
10    if (scanf("%d", &t) != 1) return 0;
11    while (t--) {
12        int n, k, p;
13        scanf("%d %d %d", &n, &k, &p);
14        int dp[1505];
15        memset(dp, 0, sizeof(dp));
16        for (int i = 0; i < n; i++) {
17            int stack_vals[35];
```

```
19        pref[0] = 0;
20        for (int j = 1; j <= k; j++) {
21            scanf("%d", &stack_vals[j]);
22            pref[j] = pref[j - 1] + stack_vals[j];
23        }
24        for (int cur_p = p; cur_p >= 0; cur_p--) {
25            for (int x = 1; x <= k && x <= cur_p; x++) {
```

```
26         dp[cur_p] = max(dp[cur_p], dp[cur_p - x] + pref[x]);
27     }
28 }
29 }
30 printf("%d\n", dp[p]);
```

```
31  }  
32  return 0;  
33}  
34
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
2
```

```
2 4 5
```

```
10 10 100 30
```

```
80 50 10 50
```

```
3 2 3
```

```
80 80
```

```
20 20
```

```
15 50
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting9.exe
```

```
250
```

```
180
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
2
```

```
2 4 5
```

```
20 20 70 30
```

```
60 30 10 50
```

```
3 2 3
```

```
60 70
```

```
15 40
```

```
20 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Sorting9.exe
```

```
200
```

```
150
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```